

Supplementary Material

A. Glossary of Key Terms

Customer Churn

The event in which a customer stops using a company's service. In this study, churn is formulated as a binary classification task (Yes / No).

Baseline Model

A simple reference model used to establish a minimum performance benchmark. The baseline in this project predicts the majority class ("No churn") for all customers.

Class Imbalance

A dataset characteristic where one class is significantly more frequent than the other. In the Telco dataset, approximately 73.5% of customers do not churn.

Naive Bayes Classifier

A probabilistic classifier based on Bayes' theorem, assuming conditional independence between features given the class label.

Decision Tree Classifier

A supervised learning model that predicts outcomes by learning hierarchical decision rules through recursive feature splitting.

Cross-Validation (CV)

A resampling technique used to estimate model performance and stability. In 5-fold cross-validation, the training data is split into five subsets, each used once for validation.

Hyperparameter Tuning

The process of selecting optimal model parameters that are not learned directly from data, such as tree depth or leaf size.

ROC Curve

A plot of the true positive rate against the false positive rate across varying classification thresholds.

AUC (Area Under the Curve)

A scalar summary of the ROC curve that measures a model's ability to distinguish between classes.

Precision

The proportion of predicted churners that are actual churners.

Recall

The proportion of actual churners that are correctly identified by the model.

F1-score

The harmonic mean of precision and recall, commonly used for evaluating models on imbalanced datasets.

Precision-Recall Curve / AUPRC

Shows precision vs recall across thresholds; **AUPRC** is especially informative under class imbalance.

Information Leakage

A modelling error where information from the test set influences training or model selection, leading to overly optimistic results.

B. Intermediate Results and Additional Analysis

B1) Decision Tree before vs after tuning

In the initial 5-fold CV run, the Decision Tree used a fixed setting (MinLeafSize = 10) and achieved:

- CV Mean AUC = 0.808, CV Mean Accuracy = 0.765

After grid-search tuning, the best configuration was:

- MinLeafSize = 2, MaxNumSplits = 50
- CV Mean AUC = 0.826 (improved separation ability)

This is the strongest “intermediate result” because it clearly shows why tuning mattered for this model.

B2) Precision–Recall behaviour (important under imbalance)

On the test set, the Precision–Recall curves show Naive Bayes maintaining higher AUPRC than the

Decision Tree:

- Naive Bayes AUPRC ≈ 0.630
- Decision Tree AUPRC ≈ 0.578

This supports what we saw in metrics: Naive Bayes is more recall-heavy, so it catches more churners but with lower precision.

B3) Generalisation check (CV vs Test AUC plot)

The “CV Mean vs Test AUC” bar chart shows CV and test AUC values are close for both models, suggesting the evaluation pipeline generalises well and isn’t overfitting to one split.

Additional Figures

B.1 Precision–Recall Curve Analysis (Figure 1)

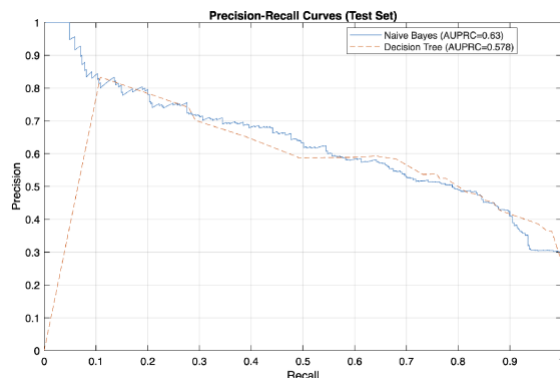


Figure S1 shows the precision–recall curves for Naive Bayes and the tuned Decision Tree evaluated on the test set. Precision–recall curves are particularly informative for imbalanced datasets, as they focus on performance for the minority class (churners).

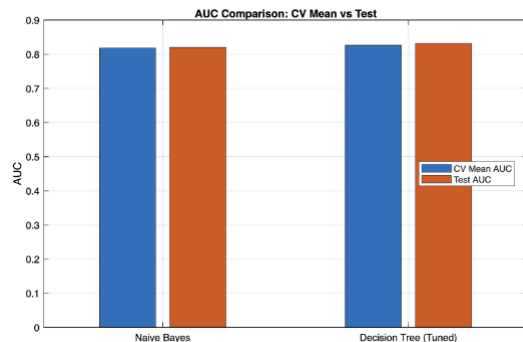
Naive Bayes achieves a higher area under the precision–recall curve (AUPRC ≈ 0.63) compared to the Decision Tree (AUPRC ≈ 0.58). This indicates that Naive Bayes maintains higher precision across a wide range of recall values and is more effective at identifying churners when recall is prioritised. In contrast, the Decision Tree shows a steeper decline in precision as recall increases, reflecting a more conservative classification behaviour.

This result complements the confusion matrix analysis and confirms that Naive Bayes is better suited to recall-focused churn detection scenarios.

B.2 Cross-Validation vs Test AUC Comparison (Figure 2)

Figure S2 compares the mean cross-validated AUC with the final test-set AUC for both models. For both Naive Bayes and the tuned Decision Tree, the CV and test AUC values are closely aligned.

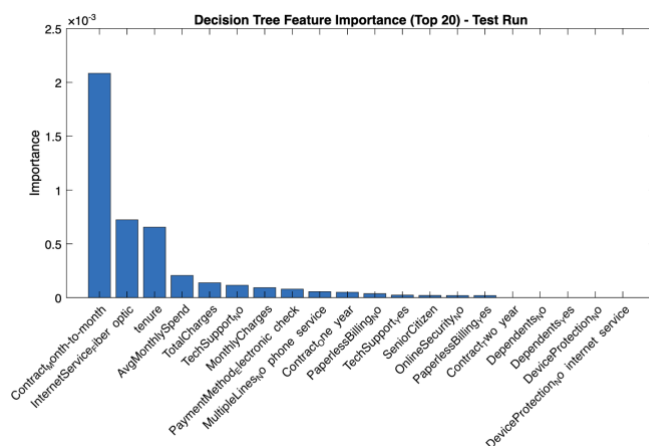
This consistency indicates that the evaluation pipeline generalises well and that no significant overfitting occurred during model selection. The tuned Decision Tree shows slightly higher AUC values in both settings, confirming that the performance gains observed during cross-validation translate reliably to unseen data.



B.3 Decision Tree Feature Importance (Figure 3)

Figure S3 presents the top 20 most important features learned by the tuned Decision Tree model. Contract-related variables (e.g. month-to-month contracts), service type (e.g. fibre optic internet), tenure, and billing-related features (average monthly spend and total charges) emerge as the most influential predictors.

This aligns with domain expectations, as customers with short-term contracts, higher monthly costs, and specific service configurations are more likely to churn. The feature importance analysis also supports the interpretability advantage of Decision Trees, providing transparent insight into which customer attributes drive churn predictions.



Several intermediate experiments informed the final modelling choices:

- The majority-class baseline achieved high accuracy due to dataset imbalance but failed to identify any churners, reinforcing the need for metrics beyond accuracy.
- An untuned Decision Tree evaluated via cross-validation achieved a lower mean AUC (≈ 0.808) compared to the tuned model (≈ 0.826), motivating hyperparameter optimisation.
- Hyperparameter tuning was not applied to Naive Bayes, as exploratory experiments showed negligible performance variation due to its fixed probabilistic structure.

These results highlight the importance of systematic evaluation and model selection rather than relying on default configurations.

D. Implementation Details and Design Choices

All experiments were implemented in MATLAB, using functions and workflows covered in the module. Core functionality relied on the *Statistics and Machine Learning Toolbox*, particularly `fitcnb` for Naive Bayes, `fitctree` for Decision Trees, `cvpartition` for data splitting and cross-validation, and `perfcurve` for ROC and precision–recall analysis. No external libraries or third-party toolboxes were used.

A structured machine learning pipeline was followed to ensure reproducibility and fair comparison. The dataset was first cleaned and preprocessed, including removal of identifiers, conversion of categorical variables, and handling of missing values. Feature engineering was deliberately kept simple and interpretable, adding only domain-relevant features such as average monthly spend and a long-term contract indicator.

Categorical variables were encoded using dummy variable expansion, implemented via a custom design matrix function to ensure consistent feature alignment between training and test sets. This explicit handling avoids reliance on hidden preprocessing steps and ensures transparency in the modelling pipeline.

A strict train–validation–test separation was enforced throughout. Cross-validation was applied only to the training set for model selection and hyperparameter tuning, while the test set was used exactly once for final evaluation. For Naive Bayes, feature standardisation was applied *within each cross-validation fold* and using training-set statistics only, preventing information leakage. Standardisation was not applied to the Decision Tree, as tree-based models are scale-invariant.

Hyperparameter tuning was performed exclusively for the Decision Tree, reflecting its sensitivity to model complexity. A grid search over `MinLeafSize` and `MaxNumSplits` was conducted using cross-validated AUC as the primary selection criterion. In contrast, Naive Bayes was left untuned due to its fixed probabilistic structure and limited hyperparameter flexibility, as confirmed by preliminary experimentation.

Evaluation metrics were chosen to reflect both overall discrimination and minority-class performance in an imbalanced setting. Accuracy, F1-score, ROC-AUC, and precision–recall curves were used together to provide a balanced and informative assessment of model behaviour.